McMaster University ENGINEERING | mohawk | W BOOTH SCHOOL OF ENGINEERING PRACTICE AND TECHNOLOGY
McMaster-Mohawk Bachelor of Technology Partnership

Laboratory 1
Introduction to LabVIEW

SMRTTECH 3CC3

Cloud Computing and Internet of Things

B. Tech - SMRTTECH

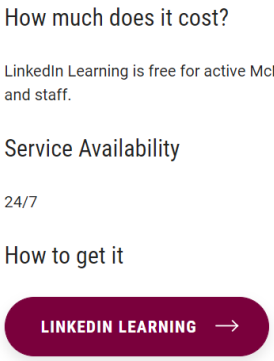
OBJECTIVES

In this lab you will learn the following features about LabVIEW Software:

1. Fundamentals of LabVIEW Graphical Programming language.
2. Controls, Indicators, VI, and SubVI.
3. Data types, arrays and clusters.
4. While loops, for loops and shift registers.

Requirements

1. An active account on Linkedin Learning. All McMaster Students should have a free access to all linkedin learning.
    a. Please visit https://uts.mcmaster.ca/services/teaching-and-learning/linkedin-learning/
    b. Click on **LINKEDIN LEARNING**, and login using your McMaster credentials.



    c. Linkedin Learning should launch.

2. LabVIEW Software is installed on your machine. See the installation guide that is posted on avenue to learn.

PROCEDURE

1. Go to https://www.linkedin.com/learning/learning-labview
2. Watch all the videos for **sections 1, 2, 3, and 4**.

EXERCISE 1

A voltage divider (Figure 1) is a very simple DC circuit that is often used to generate a desired voltage (Vout) when another voltage (Vin) is available.

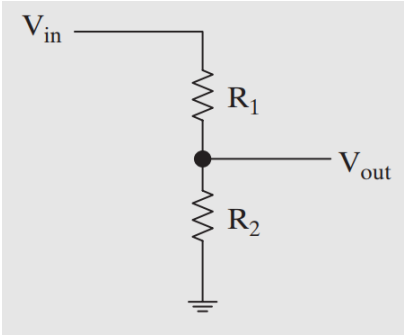


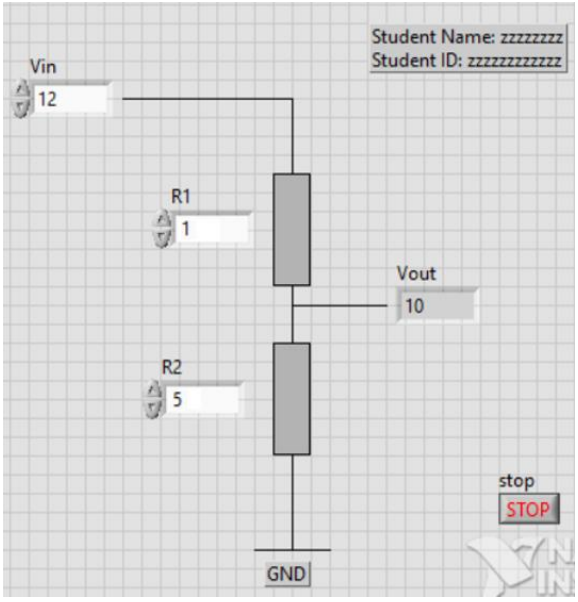
Figure 1

The math behind a voltage divider is simple:

$$V_{out} = V_{in}\left(\frac{R_2}{R_1 + R_2}\right)$$

Create a VI as shown in Figure 2 that calculates V_{out} given V_{in}, R_1 and R_2. **You must include your student name and ID in the front panel**.

Hints: Use a while loop in your block diagram, use Decorations in your front panel to add lines, boxes and labels.



EXERCISE 2

Complete the Lab 1 Quiz on Avenue to Learn.

SUBMISSION

1. Submit your VI program file. Your VI front panel must contain Student name and ID.
2. Complete Lab 1 quiz on avenue to learn